

OMID ZANGANEH

Senior GIS Developer and Analyst

MS Geography — Geographic Information Science & Technology · Workflow Automation · AI/ML Integration
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PROFESSIONAL SUMMARY

GIS Developer and software engineer specializing in GIS workflow automation, ArcGIS Pro development, and AI-powered spatial workflows. Builds production desktop applications (Python, C#), ArcGIS Pro add-ins and geoprocessing toolboxes, SQL Server data pipelines, and full-stack web tools (Next.js, React, TypeScript). Known for compressing multi-day manual processes into automated pipelines. 2026 Edison Award winner for demonstrating superior technical ability. MS Geography (GIS&T), 4.0 GPA.

WORK EXPERIENCE

Olsson

March 2025 – Present

GIS Developer

Lincoln, Nebraska ♦ 2025 Nominee & 2026 Edison Award Winner

- Architected production Python and C# desktop applications for GIS workflows including automated bore profile generation that cut processing time from days to minutes.
- Built an ArcGIS Automation Suite of custom Python geoprocessing toolboxes reducing manual GIS steps by ~90% and accelerating fiber network design timelines.
- Developed ArcGIS Pro add-ins (.NET/C#/WPF): multi-source GIS Data Downloader (OSM, USGS, FEMA, Census/TIGER, BSL), RF Analysis panel, Street View map tool, and FTTH network design dock pane.
- Engineered AI-powered tools using Azure AI Foundry, Azure OpenAI, and Google AI Studio, including RFP Radar for intelligent web-grounded contract search and classification, cutting strategic sourcing timelines from months to hours.
- Built AI object detection applications for remote inventory of utility poles, streetlights, and telecom infrastructure.

University of Nebraska at Omaha

January 2024 – August 2025

Graduate Teaching Assistant — Instructor of Record

Omaha, Nebraska

- Taught Human-Environment Geography lab sections as sole instructor of record to 150+ students across three semesters, emphasizing hands-on spatial analysis and real-world applications.

University of Nebraska at Omaha

June 2024 – August 2025

GIS Technician — Omaha Spatial Justice Project

Omaha, Nebraska

- Digitized historical land parcels from archival documents and aerial photography; reviewed legal records to build an accurate geodatabase of racially restrictive covenants in Douglas County, supporting urban spatial justice research.

SELECTED PROJECTS

RFP Radar — AI-Powered RFP Sourcing

Azure OpenAI · Google Gemini · Playwright · Python/C#

Production desktop app for intelligent, web-grounded RFP search, classification, batch county search, and CSV export. Reduced sourcing time from months to hours.

Bore Profile Automation — Directional Drilling Profile Generator

Python · C# · ArcGIS Pro · SQL Server · Matplotlib

Fully automated app that reads spatial waypoints, processes elevation models, and generates 2D/3D bore profiles. Cut processing from days to minutes.

ArcGIS Data Downloader Add-in — Multi-Source GIS Data Acquisition

C# · .NET 8 · ArcGIS Pro SDK · WPF · REST APIs

ArcGIS Pro dock-pane add-in that downloads OSM, USGS, FEMA, Census/TIGER, Wikipedia, and BSL layers directly into projects.

Fiber Automatic Expansion — Automated Fiber Build-Area Planning

C# · .NET · ArcGIS Pro SDK · WPF · Spatial Analysis

ArcGIS Pro decision-support add-in that grids the study area for density analysis, flood-fills viable build zones against configurable PPM thresholds, and generates economic scorecards with live human-in-the-loop refinement. Cut expansion planning from days/weeks to minutes.

FTTH Network Designer — Automated Fiber Optic Network Planning

C# · .NET 8 · ArcGIS Pro SDK · WPF · Kruskal MST

ArcGIS Pro add-in automating FTTH layout in three steps: place shafts at road intersections, connect homes to nearest shaft (tagging Street vs Driveway Drop by geodesic length), and run Kruskal's MST for an optimal Main Trunk / Terminal Branch backbone. Reduced network layout from days to minutes.

RF Analysis Tool — 8-Tool Wireless Planning Panel

C# · ArcGIS Pro SDK · Python

ArcGIS Pro side panel with coverage prediction, PCI/RSI planner, tilt/azimuth optimizers, interference analysis, and tower placement optimizer — all as map layers.

Aerial & Streetview AI Detection — YOLO Utility Infrastructure Detection

Python · YOLO · OpenCV · Aerial/Street Imagery APIs

Desktop apps fetching aerial tiles or traversing street routes, running custom YOLO models to detect/classify utility assets and export georeferenced results to ArcGIS.

GeoPipe — Enterprise GIS ETL to SQL Server

Python · CustomTkinter · pyodbc · SQL Server · PyInstaller

GUI ETL tool for large spatial/tabular imports into SQL Server with schema auto-detection, GEOMETRY/GEOGRAPHY support, and connection-loss auto-resume.

omidzanganeh.com — Full-Stack Portfolio & Browser GIS Tools

Next.js 16 · React · TypeScript · Leaflet · Vercel

Personal site with 10+ browser GIS tools and general web apps (geocoder, isochrone, elevation profile, census, coordinate converter, Gymflow, Stock Screener) and an AI news aggregator.

EDUCATION

Master of Science, Geography — Geographic Information Science & Technology August 2025

University of Nebraska at Omaha, Nebraska **GPA: 4.00** · **GRACA Project Award**

Thesis: Spatiotemporal Analysis of NOx Emissions from U.S. Cement Plants Using TROPOMI Data — remote sensing, hotspot analysis, environmental visualization, population exposure & environmental justice.

Bachelor of Science, Geomatics (Surveying) Engineering

August 2016

Geomatics College of National Cartographic Center (GCNCC), Tehran

TECHNICAL SKILLS

Languages & Frameworks Python · C# / .NET · SQL · TypeScript · JavaScript · HTML/CSS · React · Next.js · WPF

GIS & Spatial ArcGIS Pro · ArcGIS Pro SDK · ArcPy · Python Toolboxes (.pyt) · ArcGIS Online · ArcGIS Enterprise · QGIS · Geoprocessing · Network Analysis · Remote Sensing · Google Earth Engine

AI / ML Azure OpenAI · Azure AI Foundry · Google AI Studio / Gemini · YOLO · OpenCV · Prompt Engineering · Web Grounding · Batch Classification Pipelines

Data & Backend SQL Server · Supabase · PostgreSQL · pyodbc · ETL Pipelines · Spatial Types (GEOMETRY/GEOGRAPHY) · Smartsheet API

Web & Cloud Next.js App Router · Vite · Leaflet · Vercel · REST APIs · PWA · Microsoft Azure · Google Cloud

Desktop & Tooling CustomTkinter · PyInstaller · Playwright · Matplotlib · AutoCAD · ENVI · SNAP · Photomod · Tableau · Adobe Photoshop · Adobe Illustrator

Languages English (Fluent) · Persian (Native)

For project write-ups, live GIS tools, and demos: omidzanganeh.com
